MLOps PROJECT / INDIA / DESIGN POSTER

DELHI AQI FORECASTING

Tomorrow's air quality.

A reproducible forecast, monitored after deployment.

PRACTICAL PURPOSE

Predict next-day city AQI to support awareness and planning around pollution trends.

REAL DATA / EASY ACCESS

3,143 daily Delhi records

May 2015 - Dec 2023 / CPCB bulletin compilation
Public CSV / no account or API key

01 ML PIPELINE + CI/CD

Poster criterion 1 / CO4

01 CPCB CSV → Version + validate
02 Lag features → Seven prior AQI days
03 Train + compare → Persistence vs forest
04 MLflow tracking → Params + run metrics
05 Quality gate → Chronological holdout
06 GitHub Actions → Checks on commits
07 Docker build → Pinned dependencies
08 Artifact Registry → Versioned images
09 Cloud Run API → Candidate + smoke test
10 Dashboard → Forecast + model ID

Cloud Storage: versioned data + experiment artifacts. Promotion: checks + recorded approval.
Time order preserved / tune on validation / final test reported separately / no future AQI features

02 DEPLOYMENT STRATEGY

Docker + managed scaling

SCALE

Stateless FastAPI on Cloud Run. Scale to zero; service cap 3; tagged candidates add capacity.

RELEASE + ROLLBACK

Deploy candidate revision, smoke-test, then approve traffic switch. Restore prior revision if needed.

DEPLOYMENT CHALLENGES

Cold starts: small image. Model mismatch: pinned digest. Access: OIDC and least-privilege IAM.

03 MONITOR + RETRAIN

CO3 / feedback after actuals arrive

Forecast → Log model version
Actual AQI → Join by target date
Error + drift → MAE / input shift
Review candidate → Retrain + rerun gate

Cloud Logging: latency + failures. Drift prompts review; models never auto-promote.

04 GOVERNANCE + RESPONSIBLE AI

Poster criterion 2 / CO4

VERSIONING

Data SHA-256, Git SHA, MLflow run ID and container digest.

AUDIT TRAIL

Metrics, timestamp, named approver and deployed revision recorded.

COMPLIANCE

Public aggregate data; source attribution; IAM; log retention policy.

BIAS MITIGATION

Report seasonal errors; inspect extreme-AQI failures; disclose coverage gaps.

DEMO: historical replay > forecast > inject drift > review > deploy > rollback

Scope: city-level historical backtesting. Live forecasts need fresh inputs; no health or accuracy guarantees.

Data: UrbanEmissions / CPCB bulletins (Zenodo 11146316). Architecture: Cloud Run + MLflow documentation.